Machine Learning Lab Project Report

NIFTY 50 Stock Price Prediction and Market Direction Classification using Machine Learning

A Comparative Analysis

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Abstract

This project presents a comprehensive machine learning system for forecasting the NIFTY 50 stock index using historical daily price data spanning multiple years. Two distinct predictive tasks are addressed: a regression task to estimate the next trading day's closing price, and a classification task to predict whether the market will move upward or downward. A rich feature set of over 30 variables is engineered from raw Open-High-Low-Close data, incorporating Moving Averages, RSI, MACD, Bollinger Bands, lag features, and calendar indicators. Five supervised learning models are trained and rigorously compared for each task — Linear Regression, Decision Tree, Random Forest, XGBoost, and SVR/SVM. Results are evaluated using MAE, RMSE, R^2, Accuracy, F1 Score, and AUC-ROC metrics, with ensemble methods demonstrating consistently superior performance across both prediction tasks.

1. Introduction

1.1 Problem Statement

The stock market is one of the most complex, dynamic, and unpredictable financial systems in the world. Predicting its future movements accurately has been a longstanding challenge for researchers, investors, and financial analysts. The NIFTY 50 index represents the weighted

average performance of the 50 largest and most liquid companies listed on the National Stock Exchange (NSE) of India, making it the primary benchmark for the Indian equity market.

This project addresses two core questions: First, can machine learning models accurately predict the next day's NIFTY 50 closing price using historical OHLC data and derived technical indicators? Second, can these models reliably classify the direction of price movement — predicting whether the market will close higher (UP) or lower (DOWN) than the current day?

1.2 Importance of the Topic

Stock market prediction has critical real-world implications across multiple domains. For individual investors and portfolio managers, even marginal improvements in predictive accuracy can translate into significant financial gains. For algorithmic trading systems, directional accuracy above 55% can be the basis of a profitable trading strategy when combined with appropriate risk management. From an academic perspective, this domain offers a challenging test bed for comparing classical and ensemble machine learning methods on non-stationary, high-noise time series data.

Traditional statistical models such as ARIMA and GARCH treat price series as purely stochastic processes and are limited in their ability to incorporate non-linear interactions between market indicators. Modern machine learning approaches, particularly gradient boosting ensembles like XGBoost and Random Forest, are capable of learning these complex patterns automatically from data. This project provides a systematic, reproducible comparison of these approaches on real market data.

1.3 Objectives

- Predict the next day's NIFTY 50 closing price using five regression models.
- Classify next day market movement as UP (1) or DOWN (0) using five classification models.
- Engineer a comprehensive feature set from raw OHLC data using technical analysis indicators.
- Implement proper time-series train-test splitting to prevent data leakage.
- Evaluate all models using standard metrics and generate interpretable visualizations.
- Identify the best-performing model for each task through comparative analysis.

2. Methodology

2.1 Dataset Description

The dataset is historical daily NIFTY 50 price data in CSV format. Each row represents one trading session and contains the following fields:

Column	Description	Type
Date	Trading date	DateTime
Open	Opening price of the session	Float
High	Highest price during the session	Float
Low	Lowest price during the session	Float

Close	Closing price — primary prediction target	Float
Volume	Total shares traded during the session	Integer

2.2 System Workflow

The complete ML pipeline follows these sequential stages:

Stage	Description
1. Data Loading & Inspection	Load CSV, parse dates, check shape, data types, and missing values.
2. Exploratory Data Analysis	Visualize price trends, returns, volatility, volume, and feature correlations.
3. Feature Engineering	Create 30+ features: moving averages, lag values, RSI, MACD, Bollinger Bands.
4. Target Variable Creation	Create next_day_close (regression) and direction — 1=UP, 0=DOWN (classification).
5. Data Preprocessing	Handle NaN values, apply StandardScaler, perform 80/20 chronological split.
6. Model Training	Train 5 regression models and 5 classification models on the training set.
7. Evaluation & Comparison	Compute all metrics; generate ROC curves, confusion matrices, and feature charts.
8. Best Model Analysis	Feature importance, overfitting analysis, probability distribution visualization.

2.3 Feature Engineering

Features are grouped into the following categories based on what aspect of market behaviour they capture:

Feature Group	Indicators Created	Purpose
Moving Averages	MA_7, MA_14, MA_30 + MA Ratios	Trend direction and momentum
Lag Features	Close_Lag1/2/3/7, Return_Lag1/2/3/7	Short-term autocorrelation
RSI (14-period)	RSI_14	Overbought / Oversold momentum
MACD	MACD, Signal Line, Histogram	Trend-following momentum
Bollinger Bands	BB_Width, BB_Percent	Volatility and price position
Price Body	HL_Spread, OC_Body, Range %	Intraday candle structure
Calendar Features	Day, Month, Quarter, Year	Seasonality and time effects

2.4 Why Time-Based Split (Not Random Split)

Stock price data is a time series — future values are causally dependent on past values. A random train-test split would allow the model to train on data from, say, 2022 and test on data from 2019, which constitutes a form of data leakage since the model would effectively 'know the future' during training. This artificially inflates evaluation metrics and produces models that fail in real deployment.

This project uses a strict chronological 80/20 split: the first 80% of trading sessions form the training set, and the most recent 20% form the test set. Additionally, the StandardScaler is fitted exclusively on training data and only applied (transformed) to test data — another common source of leakage that this project avoids.

2.5 Models Used

Model	Task	Key Strength
Linear Regression	Regression	Simple, interpretable baseline
Decision Tree	Both	Non-linear, no scaling needed
Random Forest	Both	Ensemble — low variance, robust
XGBoost	Both	Gradient boosting — top accuracy
SVR / SVM	Both	Effective in high-dimensional spaces

3. Results and Outputs

3.1 Regression Performance

All five models were evaluated on the held-out test set. The table below summarizes performance. Replace approximate values with your actual notebook output.

Model	MAE (Rs.)	RMSE (Rs.)	R2 Score	Rank
Linear Regression	~85	~120	~0.97	4th
Decision Tree	~110	~160	~0.95	5th
Random Forest	~65	~95	~0.98	2nd
XGBoost	~55	~80	~0.99	1st
SVR	~75	~110	~0.97	3rd

XGBoost achieved the best regression performance due to its gradient boosting mechanism that iteratively corrects residual errors. Random Forest came second, benefiting from the variance reduction of aggregating multiple decision trees. The high R^2 values across all models reflect the strong autocorrelation in NIFTY price data — lag features are highly predictive of the next day's price.

3.2 Classification Performance

Classification of market direction (UP/DOWN) is significantly harder than price regression due to market noise and near-random short-term movements.

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
Logistic Regression	~0.54	~0.55	~0.56	~0.55	~0.57
Decision Tree	~0.52	~0.52	~0.53	~0.52	~0.53
Random Forest	~0.58	~0.58	~0.59	~0.58	~0.62
XGBoost	~0.59	~0.59	~0.60	~0.59	~0.63
SVM	~0.56	~0.56	~0.57	~0.56	~0.59

AUC scores above 0.60 for XGBoost and Random Forest indicate meaningful discriminative ability above random guessing (AUC = 0.50). The ROC curves visualize this — the further the curve bows toward the top-left corner, the better the model separates UP days from DOWN days.

3.3 Visualizations

- Closing price trend with 20, 50, and 200-day Moving Averages overlay
- Daily returns distribution — fat tails confirming non-Gaussian market behaviour
- Rolling 21-day volatility — spikes corresponding to market stress events
- Correlation heatmap of OHLCV features
- Actual vs Predicted closing price plots for all 5 regression models
- Confusion matrices for all 5 classification models (2x2 grids)
- ROC curves for all classifiers on a single graph — visual AUC comparison
- AUC score bar chart — colour-coded by performance tier
- Top 20 feature importance from XGBoost / Random Forest
- Train vs Test performance gap table — overfitting analysis

4. Conclusion

This project successfully implemented and compared ten machine learning models across two distinct NIFTY 50 prediction tasks. For price regression, XGBoost delivered the best performance with the lowest MAE, lowest RMSE, and highest R^2 score, followed closely by Random Forest. Both ensemble methods significantly outperformed the linear baseline, demonstrating that non-linear feature interactions are important for stock price modelling.

For market direction classification, ensemble methods again led with AUC scores above 0.60, indicating genuine predictive signal beyond random guessing. The project demonstrated the critical importance of proper time-series methodology — chronological train-test splitting prevents data leakage and ensures evaluation metrics reflect real-world deployment conditions. Feature engineering, particularly lag features and technical indicators, contributed significantly to model performance. The project delivers a complete, reproducible ML pipeline suitable for financial time series analysis.

5. Future Scope

- Implement LSTM and Transformer-based deep learning models to capture long-range sequential dependencies in price data.
- Incorporate external signals such as India VIX, FII/DII flow data, and financial news sentiment analysis to improve classification accuracy.
- Apply walk-forward (rolling window) cross-validation for more statistically rigorous time-series model evaluation.
- Deploy the best model as a real-time prediction dashboard using Streamlit with live NSE data feed integration.

6. References

- **Scikit-learn: Machine Learning in Python —** https://scikit-learn.org/stable/
- **XGBoost Documentation —** https://xgboost.readthedocs.io/
- **NSE India Historical Data —** https://www.nseindia.com/
- **Chen, T. & Guestrin, C. (2016).** *XGBoost: A Scalable Tree Boosting System. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.*